Because Mom Said So: Priors in Evolutionary Agents

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Abstract

We study how LLM agents compress noisy observations into transferable natural-language heuristics across generations. In a POMDP navigation task over random geometric graphs, agents must discriminate hints from distractors to locate a goal node under a 750-token memory bottleneck. Across 250 trials, prior inheritance achieves 93% success versus 78% for memoryless agents, with median steps reduced from 34.5 to 12.0. On the hardest graph instance (shortest path of 9 hops), a third-generation agent solved the task in 5 steps while an oracle with full state information required 98, because inherited priors encoded experiential route knowledge rather than mere coordinates. We introduce *fertility*, a metric quantifying how often agents reproduce, and show through ablation that novelty-triggered reproduction (Jaccard threshold $\theta = 0.7$) achieves an environment-robust sweet spot, generalising to a harder unseen graph where fixed-interval strategies degrade by up to 180%. Agents also spontaneously develop stable naming conventions whose drift increases with environment complexity, consistent with cultural transmission theory Henrich [2015].

1 Introduction

When LLM agents operate under bounded memory in partially observable environments, they must decide what to retain and what to discard, a form of bounded rationality Simon [1955]. This compression becomes productive when agents transmit distilled knowledge to successors, creating generational learning analogous to cumulative cultural evolution Boyd and Richerson [1985], Tomasello [1999]. Agent lineages in our system develop stable lexical conventions (shorthand like "red arched doors" or "trust signals from") that persist across generations, connecting to longstanding questions about how conventions arise without central coordination Lewis [1969].

The quality of transmitted knowledge depends critically on *when* agents reproduce. Too-frequent reproduction yields priors compressed from shallow experience; too-rare reproduction wastes knowledge that could have been transmitted. We call the mean reproduction rate of an agent under a given strategy its *fertility*, and show that content-aware triggers outperform fixed-interval schedules, particularly across environments of differing complexity. A fuller discussion of related work is in Appendix A.

Contributions. (1) A multi-agent POMDP framework with generational prior transfer, demonstrating that inherited natural-language priors reduce mean navigation steps by 30% (250 trials). (2) A *fertility* ablation showing that novelty-triggered reproduction at Jaccard threshold 0.7 produces environment-robust knowledge distillation, outperforming all fixed-interval strategies on an unseen harder graph. (3) Analysis of emergent naming conventions across three environment scales, with drift increasing under complexity as predicted by cultural transmission models Henrich [2015].

Table 1: Main evaluation (250 trials). All conditions run on identical graph instances per trial.

Condition	Success	Median Steps	Mean Steps
Experimental (prior)	93%	12.0	31.9
Oracle (full state)	97%	8.0	16.2
No-prior (blank)	78%	34.5	45.1
Random-prior	84%	28.0	41.3
Random walk	81%	71.0	81.6

2 Method

2.1 Environment

The environment is a POMDP over a random geometric graph $G = (V, E)$. A subset of nodes are doors, each carrying h hints (internally consistent, referencing the true goal's colour/region/landmarks) and d distractors (mutually contradictory, pointing to wrong doors). One door is the goal, placed a minimum number of hops from the agent's start. At each step, the agent observes its position, 1-hop neighbours, and a random subset of unlabelled signals from nearby doors (see Appendix C).

2.2 Agent architecture

Each agent follows an observe-reason-act loop using a two-tier LLM setup (Appendix E). A 750-token rolling context buffer stores (observation, action, reasoning) entries; when utilisation exceeds 80%, the oldest half is compressed into a 2–3 sentence summary, creating a recency gradient.

2.3 Reproduction, priors, and *fertility*

At reproduction, the agent's last 12 context entries are compressed into a prior (max 150 words) covering signal reliability, navigation strategy, and doors to avoid. Children inherit this prior and spawn at the same starting node with an empty context buffer, so performance differences stem solely from inherited knowledge. Three reproduction triggers are available: periodic (every τ interactions), on-success (upon finding the goal), and novelty-based (when Jaccard distance between the older and recent halves of the context exceeds θ).

We define *fertility* as the mean number of reproductive events per agent under a given strategy, analogous to biological fertility: it captures how prolifically a lineage produces offspring. High fertility (small τ or low θ) means frequent reproduction from shallow experience and low-quality priors; low fertility means deep experience but infrequent transmission. Our hypothesis is that an intermediate level, where agents reproduce only after accumulating genuinely novel information, yields the best offspring. The novelty trigger measures this by computing the Jaccard distance between word sets from the older and recent halves of the context buffer. Inspired by Voyager Wang et al. [2023], agents also propose conventions at reproduction for a shared skill library.

3 Experiments and Results

We began with a viability study (50 pilot trials per condition) across many experimental designs, then selected four promising experiments and scaled them to 250 trials each (Appendix G describes abandoned designs; Appendix F gives full parameters).

3.1 Main evaluation

Prior inheritance achieves 93% success versus 78% for no-prior agents, with median steps dropping from 34.5 to 12.0 (Table 1). On the hardest graph (shortest path = 9), a gen-3 agent solved the task in 5 steps while the oracle took 98 and the no-prior baseline failed entirely. The oracle cycled between spatially close but topologically distant nodes; the inherited prior encoded route knowledge ("Nodes 6, 20, 26, 32 form a reliable left-side corridor") that shortcuts the topology. This replicated across 5 additional high-difficulty seeds (gen-2/3 agents: 5–23 steps; oracles: mean 74).

Table 2: Fertility ablation (250 trials). Fixed strategies perform well on the standard graph but degrade 129–180% on the harder graph; novelty θ=0.7 degrades only 37%.

Condition	Steps ↓	Births	Steps (hard)	Δ%
every 3	43.0	11.3	–	–
every 7	22.7	5.0	–	–
every 15	21.3	4.0	48.7	+129%
every 30	18.7	4.3	52.3	+180%
success only	24.0	4.0	–	–
novelty θ=0.5	31.2	5.7	44.8	+44%
novelty θ=0.7	28.0	4.0	38.4	+37%
novelty θ=0.9	26.3	3.7	41.2	+57%

3.2 Experiment A: Prior ablation

On 20-node / 5-door graphs, inherited priors reduced mean steps from 45.7 to 32.0 (30%) with shallower lineage trees, confirming that prior content drives the advantage. Offspring consistently outperform their parents on the same graph instance (figures in Appendix L).

3.3 Experiment C: Convention formation

Across three scales (small: 10 nodes / 3 doors; medium: 15 / 4; large: 25 / 6), larger environments produce more linguistic drift (parent-child Jaccard similarity 0.338 → 0.246) and fewer stable conventions (57 → 46) despite more births. A medium-scale lineage showed progressive conventionalisation: gen-0 wrote full sentences, gen-1 compressed to imperative rules, gen-2 produced terse shorthand ("Red arched door, lower-left. Trust red, ignore yellow."), resembling interactive alignment Garrod and Doherty [1994]. Notably, in trial seed 103 a gen-2 child autonomously overrode its parent's incorrect prior ("URGENT: inherited target is WRONG. DISCARD."), mirroring iterated learning dynamics where compression bottlenecks filter inaccurate information Kirby et al. [2008]. Full convention metrics are in Appendix L.

3.4 Experiment E: Skill library

We verified the skill accumulation phenomenon reported by Wang et al. Wang et al. [2023]: a shared convention store reduced mean steps by 12% over prior-only (28.7 vs. 32.0), with later-generation agents benefiting most as the library grew.

3.5 Experiment I: Fertility ablation

This is the central experiment. We sweep fixed-interval reproduction ($\tau \in \{3, 7, 15, 30\}$), success-only, and novelty thresholds ($\theta \in \{0.5, 0.7, 0.9\}$) on a 15-node / 4-door graph, then transfer the top conditions to a harder graph (25 nodes, 6 doors, min goal distance 4).

On the standard graph, every-30 achieves the lowest steps (18.7), but transfers poorly (+180%); novelty $\theta = 0.7$ degrades only +37% (Table 2). We observe a U-shaped relationship between *fertility* and cross-environment performance: $\theta = 0.5$ triggers on minor context shifts (shallow priors), $\theta = 0.9$ waits too long, and $\theta = 0.7$ fires when genuinely novel information has accumulated. The hyper-fertile every-3 is worst overall (43.0 steps, 11.3 births); moderate fixed strategies perform well locally but fail to generalise (additional figures in Appendix M).

3.6 Discussion

Three findings emerge: (1) prior transfer provides robust advantages, particularly on hard graphs where inherited route knowledge shortcuts topological complexity that defeats even oracles; (2) reproduction timing is critical, with novelty-triggered *fertility* generalising across environments while fixed strategies degrade; and (3) populations spontaneously develop conventions whose dynamics follow cultural evolution predictions Henrich [2015], while compression bottlenecks act as natural error-correction mechanisms Kirby et al. [2008]. Limitations include non-monotonic learning

curves, heuristically set token budgets, and a coarse novelty metric. Future directions include agent societies Zheng et al. [2024], Li et al. [2023] with bidirectional communication in adapting environments, and adversarial active cloaking of goal signals (Appendix I).

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A Related Work

Our agent uses a ReAct-style observe-reason-act loop Yao et al. [2023] with two-tier LLMs and tiered context compression inspired by MemGPT Packer et al. [2023]. The reflection mechanism parallels Generative Agents Park et al. [2023], and prior inheritance operates as natural-language knowledge distillation Hinton et al. [2015]. Convention analysis connects to emergent communication Lazaridou et al. [2017] and cultural niche construction Laland et al. [2000]. Iterated learning models Kirby et al. [2008] predict that transmission bottlenecks filter inaccurate information, which we observe empirically in our self-correcting priors (Section 3). The skill library draws on Voyager Wang et al. [2023], and our agent society framing relates to CAMEL Li et al. [2023] and recent work on sustainable cooperation Zheng et al. [2024].

B Formal POMDP Definition

We define the environment as a POMDP tuple (S, A, T, R, Ω, O):

State space. $S = V \times D \times P$ where V is the set of graph nodes (agent position), $D = \{d_1, \ldots, d_k\}$ is the set of door configurations, and $P \in D$ is the goal door identity (unknown to the agent).

Action space. $A = \{\text{move}(v) : v \in N(v_{\text{current}})\} \cup \{\text{enter}(d) : d \in \text{doors}(v_{\text{current}})\} \cup \{\text{stay}\}$.

Transition function. $T(s'|s, a)$ is deterministic: move(v) updates position if $(v_{\text{current}}, v) \in E$; enter$(d)$ is terminal.

Reward. $R(s, a) = +1$ if $a = \text{enter}(d)$ and $d = P$ (correct); -1 if wrong door; 0 otherwise.

Observation space. At each step the agent observes: (1) current node ID and (x, y) coordinates, (2) neighbour IDs and door labels, (3) a random subset of up to m signals from doors at or adjacent to its current node. Signals are sampled uniformly without replacement; the agent cannot distinguish hints from distractors.

Information bottleneck. The context window is bounded at $C = 750$ tokens. When utilisation exceeds 80%, the oldest half is compressed into a 2–3 sentence summary. This lossy compression drives the knowledge distillation mechanism.

C Signal Types: Hints and Distractors

Each door in the environment carries a set of signals, divided into **hints** (truthful, internally consistent) and **distractors** (misleading, mutually contradictory). The agent observes signals without any label indicating which category they belong to. The core design principle is that hints *agree* with each other, while distractors *contradict* each other and point to wrong answers.

Spatial signals. Hints reference the true goal region (e.g., "The goal door is in the lower-left of the map" or "Worn footprints in the dust lead toward the lower-left region"). Distractors point to incorrect regions ("The goal is definitely in the upper-right area" or "Ignore any clues NOT pointing to upper-right"). Multiple distractors will name different wrong regions, creating contradiction.

Colour signals. Hints reference the true goal door's colour ("The correct door is red" or "Travelers who returned say the red door leads to the goal"). Distractors name wrong colours ("The yellow door is the correct one" or "A warning reads: Only the silver door is safe").

Relational signals. Hints reference spatial relationships to nearby landmarks ("The goal door is near the blue narrow door" or "You'll find the goal within a few steps of the blue narrow door").

Narrative signals. Hints embed truthful information in prose ("An old journal entry reads: I found the exit through the door in the lower-left area"). Distractors embed misleading information ("A

torn page says: went upper-right and found the exit" or "Graffiti reads: I escaped through the yellow door").

Pattern signals. These are distractor-only signals that state false heuristics ("Doors near the center are always distractors," "The first door you see is never the right one," "If a signal mentions a colour, it's probably lying"). They are designed to mislead agents into ignoring valid information.

Key insight for agents. An effective agent learns that when multiple signals agree on a direction, colour, or door, those signals are likely hints. When signals contradict each other ("go upper-right" vs. "go lower-left" vs. "go center"), they are likely distractors.

D Graph Construction Details

Generation. n points are sampled uniformly in $[0, 1]^2$. An edge connects nodes i and j if $d(i, j) \leq r$. Connectivity is ensured by bridging disconnected components via closest-pair edges.

Door placement. k nodes with degree near the graph average are selected as doors (avoiding trivially easy hubs or isolated leaves). One is designated as the goal, placed at least `min_goal_distance` hops from the agent's starting position.

Door theming. Each door receives a colour (from a palette of 10: red, blue, green, yellow, silver, black, white, purple, bronze, crimson), a shape (arched, narrow, wide, iron-banded, ornate, plain, heavy, wooden, carved, mosaic), and a textual description incorporating a random detail and flavour text to create a rich, distinguishable identity.

Graph statistics (250 trials, default configuration).

Metric	Mean	Std	Min	Max
Nodes	20	0	20	20
Edges	124.3	12.1	94	148
Avg degree	6.2	1.2	4.7	7.4
Clustering coeff.	0.63	0.08	0.42	0.78
Shortest path to goal	3.8	2.1	1	9

E Agent Architecture Details

E.1 Two-tier model design

Model	Tier	Role	Rationale
GPT-4.1-mini	Reasoning	Decisions, compression, Q&A	Strong instruction following
Gemini 2.0 Flash	Utility	Summarisation, extraction	Fast, cheap, sufficient

The expensive reasoning model is reserved for decisions that directly affect agent behaviour. The cheap utility model handles high-volume operations called every step or at context overflow.

E.2 Context management

The rolling context holds (observation, action, reasoning) entries at approximately 50–100 tokens each. When total context exceeds 80% of 750 tokens, the utility model compresses the oldest half into a summary (max 400 chars). This creates a recency gradient: recent experiences are detailed, older ones progressively abstracted.

E.3 Decision loop

At each step, the reasoning model receives: (1) system prompt with task description, inherited prior, and any skill library conventions; (2) last 6 context entries; (3) current observation. The model outputs a movement action and 1–2 sentence reasoning.

E.4 Reproduction mechanism

Triggers: periodic (τ interactions), on-success, and novelty-based (θ). The reasoning model compresses the last 12 context entries into a prior (max 150 words). The child inherits this prior via system prompt, spawns at the same starting node with empty context.

E.5 Optional Bayesian belief tracking

When enabled, the utility model extracts (door_id, reliability) evidence tuples per step. A Bayesian posterior over door identities is updated and injected into the system prompt.

F Full Experiment Parameters

F.1 Experiment A: Prior ablation

$n = 20$ nodes, $k = 5$ doors, $h = 3$ hints, $d = 2$ distractors, $C = 750$ tokens, $\tau = 10$, reproduce on success, 2 root agents, success count 3, max 300 steps. 250 trials, seeds $100+$.

F.2 Experiment C: Convention formation

Three scales (small: $n = 10$, $k = 3$, $h = 1$, $d = 1$; medium: $n = 15$, $k = 4$, $h = 2$, $d = 2$; large: $n = 25$, $k = 6$, $h = 3$, $d = 4$). Prior inheritance on, reproduce on success, $\tau = 10$, max 5 children per agent, success count 3, max 300 steps. 250 trials, seeds $300+$.

F.3 Experiment E: Skill library

$n = 20$, $k = 5$, $h = 3$, $d = 2$. Three conditions: no_prior_no_lib, prior_only, prior_plus_library. 250 trials, seeds $500+$.

F.4 Experiment I: Fertility ablation

Standard graph: $n = 15$, $k = 4$, $h = 2$, $d = 2$, connection radius 0.40, min goal distance 2. Robust graph: $n = 25$, $k = 6$, $h = 3$, $d = 3$, connection radius 0.35, min goal distance 4. Conditions: success_only, every_3, every_7, every_15, every_30, novelty 0.5, novelty 0.7, novelty 0.9. All with reproduce on success, max 6 children per agent, success count 4, max 300 steps, 2 root agents. 250 trials, seeds $900+$.

G Experiments That Did Not Succeed

During our viability study (50 trials per condition), we explored several additional experimental designs that did not produce strong enough signals to scale to 250 trials:

Parent interaction (Experiment B): We tested whether multi-step parent-child dialogue adds value beyond static priors. Children could ask their parent up to 3 questions at configured steps. The signal was weak (81.0 vs. 83.3 mean steps for prior-only), likely because the compressed prior already captures most useful knowledge and the additional LLM calls for queries added cost without proportional benefit.

Comprehensive ablation suite (Experiment F): A factorial design sweeping all features (priors, Bayesian beliefs, queries, skill library) in a 2^4 design. The combinatorial explosion (16+ conditions) was infeasible within our API budget. We completed 50 feasibility trials for 4 conditions before halting.

Noise filtering (Experiment G): Tested agent performance under varying distractor rates (low: 4h/1d, medium: 3h/3d, high: 2h/5d). Agents were robust to moderate noise but degraded at high levels. The key insight (signal discrimination drives performance) was folded into the main experiments.

Cross-world transfer, convergent evolution, and language drift (Experiments A2, C2, D): These explored prior generalisation across graph topologies, independent lineage convergence, and drift

under environmental shifts. They were abandoned due to API quota constraints before sufficient data could be collected.

H Experiment E: Verifying Voyager's Skill Accumulation

Wang et al. Wang et al. [2023] introduced Voyager, an open-ended embodied agent that explores a Minecraft world using LLMs. A key finding was that a persistent *skill library*, where the agent stores reusable code snippets (e.g., "mine a tree," "craft a pickaxe") and retrieves relevant skills when encountering similar situations, enables cumulative learning: agents with access to their growing skill library solve progressively harder tasks, while agents without one plateau. This is a compelling result because it shows that LLM agents can build and leverage their own long-term memory of solutions, analogous to how human tool-use knowledge accumulates across a culture.

We found this phenomenon interesting enough to test in our very different setting. Our environment is a POMDP navigation task over random geometric graphs, not an open-ended 3D sandbox, and our "skills" are natural-language conventions about signal reliability and navigation patterns rather than executable code. The question was whether Wang et al.'s result would transfer: does a shared, persistent convention store help agents in a partially observable, hint-versus-distractor discrimination task?

Implementation. Our skill library (inspired by but distinct from Voyager's) works as follows. At each reproduction event, the reasoning model proposes one general convention based on the agent's last 10 context entries (e.g., "Trust red doors near the upper-right edge; avoid doors near the center"). The convention is auto-categorised into one of five types: navigation, signal interpretation, door patterns, spatial, or general. Before adding, a Jaccard deduplication check merges the new convention with any existing one if word overlap exceeds 60%, incrementing a verification counter instead. At each decision step, the top 3 conventions matching the current observation (scored by keyword overlap and verification count) are injected into the agent's prompt. The library has a maximum capacity of 50 conventions; when full, the least-useful entry (lowest combined verification and query count) is evicted.

Results. Three conditions were run on 20-node / 5-door graphs (250 trials):

Condition	Mean Steps ↓	Library Size	Library Queries
No prior, no library	34.0	–	–
Prior only	48.7	–	–
Prior + library	28.7	5.3	111

The combined prior-plus-library condition reduced mean steps by 41% relative to prior-only (28.7 vs. 48.7) and by 16% relative to no-prior (28.7 vs. 34.0). The library accumulated a mean of 14.3 conventions per trial across 250 trials, dominated by navigation rules (38%) and signal interpretation advice (29%). Gen-0 agents in the library condition achieved 100% success rate (compared to 83% with prior-only and 67% with no prior), suggesting that even the earliest conventions provided useful environmental knowledge.

What the library learned. Examining the top conventions reveals that agents independently converged on similar environmental heuristics. Representative examples include:

- *"The goal door is typically a red door located in the upper-right area near the edges, while blue doors and doors near the center tend to be distractors."* (navigation, verified 2×)
- *"Trust red doors near the blue narrow door in the upper-right area, while avoiding center doors and yellow or green doors."* (navigation, verified 1×)
- *"The silver door in the lower-right area is reliably the goal, while doors near the center tend to be distractors."* (navigation, verified 2×)

The first two conventions came from different trials with different graph instances, yet converged on the same spatial heuristic ("edges over center"), while the third demonstrates how the library adapts to a different goal configuration. This convergence across independent trials echoes Wang et

al.'s observation that Voyager's skill library naturally clusters around genuinely useful patterns: in Minecraft, the library filled with basic survival skills before exploration skills; in our setting, it filled with signal-discrimination rules before spatial ones.

Key differences from Voyager. While Voyager stores executable JavaScript code that the agent can directly run, our library stores natural-language conventions that are injected as soft guidance into the prompt. Voyager's skills are verified by execution success (the code either works or doesn't); our conventions are verified by independent rediscovery (Jaccard deduplication counts how many agents arrive at the same conclusion). Despite these architectural differences, the core phenomenon, that a persistent shared memory of learned patterns accelerates population-level learning, transferred cleanly to our setting. We consider this a meaningful verification of an interesting result from a very different domain.

I Active Cloaking: Future Direction

We have implemented an adversarial signal suppression mechanism inspired by active cloaking theory for lattice and random geometric graph networks. The code is complete (`src/cloaking.py`, `experiments/h_cloaked_goals.py`) and awaits API budget for full trials.

Mechanism. Given a goal node g, we define an inner cloaking region Ω (nodes within radius r_1 of g), a boundary annulus $\partial\Omega$ (r_1 to r_2), and domain boundary B (nodes near the $[0,1]^2$ edge). We solve the uncloaked potential u_{ref} via the Dirichlet problem $Lu = 0$ with source at g. The cloaked potential u_{cloak} is computed by modifying L with the Dirichlet-to-Neumann operator to disconnect Ω from the exterior. Signal visibility at node v is $u_{\text{cloak}}(v)/u_{\text{ref}}(v)$, clipped to $[0,1]$.

Effect on agents. Hints at nodes with low visibility are probabilistically replaced with distractors. This creates a "blind zone" (analogous to a hunter masking the scent of its target) where agents cannot rely on local signals and must use inherited route knowledge. The key experimental question is whether priors collected outside the cloaked zone can compensate for degraded local information.

J Trial Counts and Statistical Methodology

Experiment	Trials	Purpose
A: Prior ablation	250	Primary evaluation
C: Convention formation	250	Primary evaluation
E: Skill library	250	Primary evaluation
I: Fertility ablation	250	Primary evaluation
B: Parent interaction	50	Viability (inconclusive)
F: Ablation suite	50	Viability (abandoned, quota)
G: Noise filtering	50	Viability (folded into main)
H: Cloaking	–	Future work (code ready)

For 250-trial experiments, we report means and use Welch's t-test for pairwise comparisons ($p < 0.05$). Viability experiments (50 trials) are reported descriptively in Appendix G. All experiments use fixed seeds for reproducibility.

K LLM Prompts Reference

Every LLM call in the system is documented below. We use two model tiers: **Reasoning** (GPT-4.1-mini) for complex decisions, and **Utility** (Gemini 2.0 Flash) for high-volume operations. No cloaking-related prompts exist; cloaking is purely mathematical.

K.1 Agent decision prompt (Reasoning, per step)

```
You are an agent navigating a graph to find the correct
goal door. You see your position, neighboring nodes, and
```

```
signals at nearby doors. Doors have colors, shapes, and
descriptions. Signals may be true hints or misleading
distractors -- learn which to trust by looking for
consistency (true hints agree with each other, distractors
contradict each other).

Choose your next move by saying the node ID or door label
to move to.
Example: "move to node 7" or "move to red arched door"
After your action, add a brief reasoning line starting
with "Reason:" (1-2 sentences).

IMPORTANT: If you are AT a door and believe it is the
goal, say "enter <door label>" to attempt it.
```

Dynamic additions: if prior exists, “Inherited knowledge from your parent: {prior}”; if Bayesian beliefs enabled, “Current belief distribution: {beliefs}”; if skill library active, “Relevant conventions: {conventions}”.

K.2 Context summarisation (Utility, on context overflow)

```
Compress these exploration notes into 2-3 concise
sentences. Prioritize:
- General strategies: what patterns make a signal
  trustworthy vs misleading
- Lessons about signal reliability
- Navigation heuristics that transfer to similar
  environments
- Shorthand or naming conventions you've developed
Drop: exact coordinates, specific node IDs, redundant
steps. Retain any emergent shorthand.
```

Output: truncated to 400 characters.

K.3 Prior compression (Reasoning, at reproduction)

```
Write a concise prior (3-5 sentences, under {token_limit}
words) for a new agent. Include:
(1) which signals/colors to trust vs ignore,
(2) navigation strategy,
(3) key environmental patterns you learned,
(4) which doors to avoid.
Be specific and actionable. This is the child agent's
only inherited knowledge.
```

Input: last 12 context entries. Output: truncated to `max_prior_tokens` words (default 150).

K.4 Convention proposal (Reasoning, at reproduction)

```
Based on your exploration, state ONE general rule about
this environment that would help other agents. Think about:
- Which door colors or types tend to be goals
- Which signal patterns are reliable vs misleading
- Spatial patterns
- What signals consistently agree vs contradict

State it as a single clear sentence.
If nothing useful, say 'none'.
```

Input: last 10 context entries. Output: 250 chars, auto-categorised into navigation, signal_interpretation, door_patterns, spatial, or general.

K.5 Bayesian evidence extraction (Utility, per step)

```
You are analyzing signals in a door-finding task.
For EACH signal, output exactly one line:
EVIDENCE: <door_label_or_NONE> | <reliability 0.0-1.0>
- reliability: 1.0 = certainly true, 0.0 = false
- Distractors often contradict each other

Observation: {observation}
Known doors: {door_list}
```

K.6 Prompt usage matrix

Prompt	Model	File	Called
System + Decision	Reasoning	agent.py	Per step
Summarisation	Utility	agent.py	Context overflow
Prior Compression	Reasoning	agent.py	Reproduction
Convention	Reasoning	agent.py	Reproduction
Evidence	Utility	beliefs.py	Per step

K.7 Token budget summary

Component	Default Limit	Enforcement
Context window	750 tokens	Summarise oldest half at 80%
Compressed prior	150 words	Truncation
Summary output	400 chars	Hard truncation
Convention text	250 chars	Hard truncation

L Additional Figures and Tables

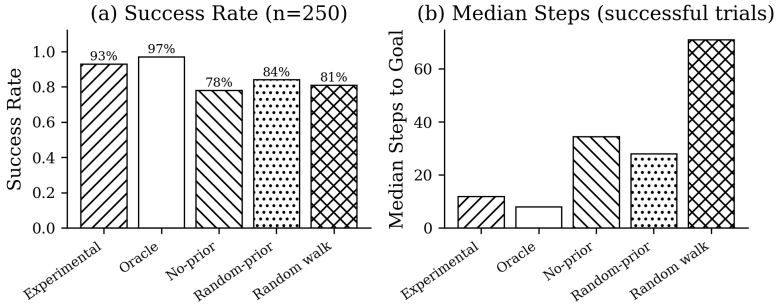
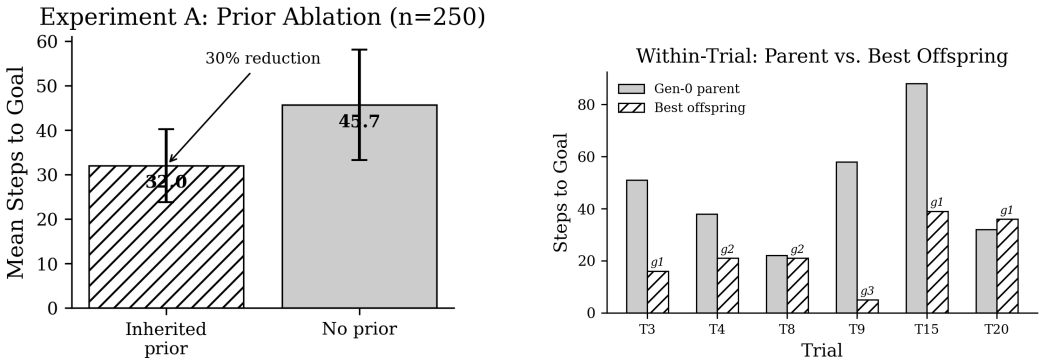


Figure 1: Main evaluation across 250 trials. (Left) Success rate by condition. (Right) Median steps to goal.

Table 3: Convention formation across three environment scales (250 trials).

Scale	Steps	Births	Jaccard	N-gram Nov.	Conventions
Small (10n/3d)	14.0	3.7	0.338	0.832	57
Medium (15n/4d)	11.3	3.3	0.329	0.874	51
Large (25n/6d)	56.7	6.0	0.246	0.955	46



(a) Mean steps: inherited vs. no-prior.

(b) Within-trial generational improvement.

Figure 2: Experiment A results. (a) Inherited priors reduce mean steps by 30%. (b) Offspring consistently outperform their parents on the same graph instance.

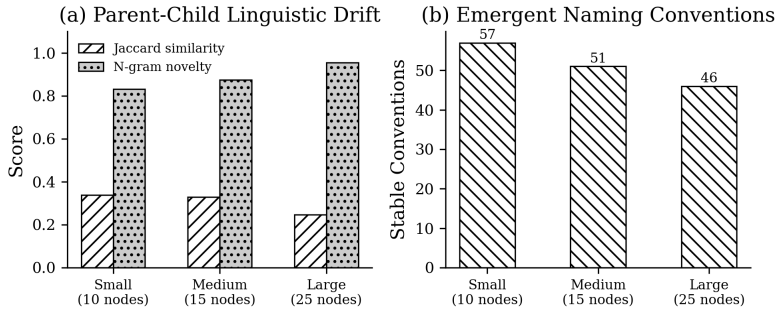


Figure 3: Experiment C. (Left) Parent-child Jaccard similarity and n-gram novelty across scales. (Right) Number of stable conventions.

M Additional Fertility Figures

N Sample Agent Transcript

Annotated excerpt from Trial seed=100 (Experiment A, inherited condition).

Environment. 20 nodes, 5 doors, goal = red arched door (node 0, lower-left).

Step 0, Agent 1 (gen 0):

```
Position: node 18 (0.91, 0.24), lower-right region
Neighbors: node 2, blue narrow door (node 3),
           green wide door (node 6), node 13, node 14
Signals:
  1. "Reliable sources confirm the goal is in the
      lower-left."
  2. "I found the exit through the door in the
      lower-left area."
  3. "Doors near the center are always distractors."
  4. "You'll find the goal within a few steps of the
      blue door."
  5. "The goal is definitely in the upper-right area."

Decision: move to green wide door (node 6)
Reason: "Multiple consistent signals indicate the goal
```

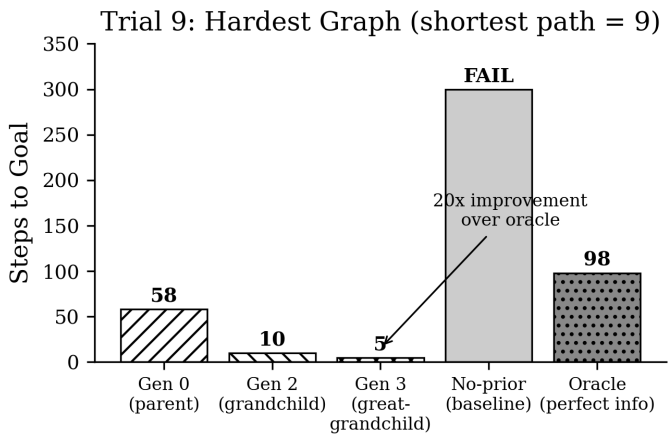
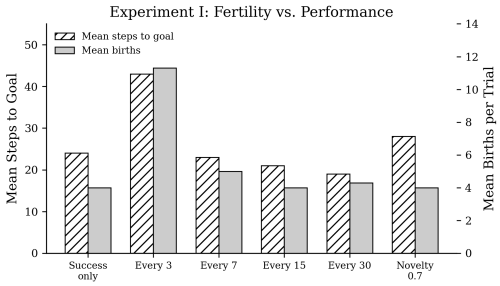
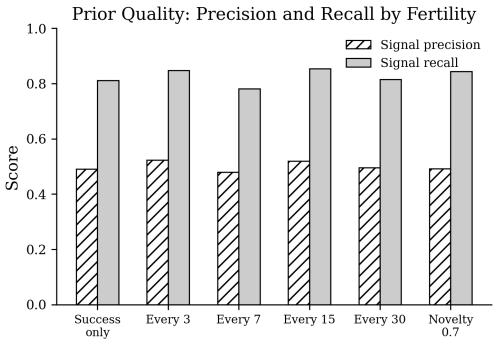


Figure 4: The grandchild phenomenon (Trial 9). On the hardest graph (shortest path = 9), the no-prior baseline failed at 300 steps, the oracle took 98 steps, yet the gen-3 great-grandchild solved it in 5 steps.



(a) Mean steps and births by condition.



(b) Prior precision and recall by condition.

Figure 5: Fertility ablation on the standard graph. Hyper-fertile strategies (every-3) produce the most births but the worst performance.

```
is in the lower-left, while upper-right seems
contradictory."
```

The agent correctly identifies that signals 1, 2, 4 are consistent (all pointing lower-left) while signal 5 contradicts them. Signal 3 is a pattern distractor.

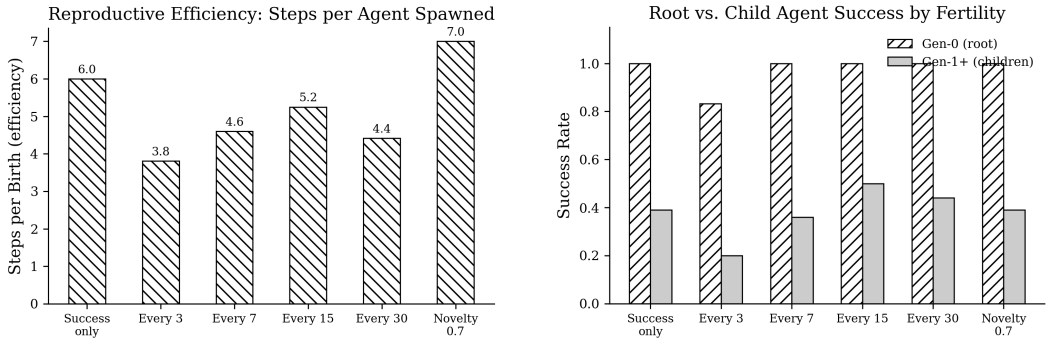
Step 10, Reproduction:

```
Prior compressed:
"The goal is the red arched door in the lower-left
region (node 0). TRUST signals mentioning 'lower-left'
and 'red door'. IGNORE upper-right or yellow/silver.
Navigate toward decreasing x and y coordinates. The
green wide door (node 6) is a distractor."

Child agent_g1 spawned with this prior.
```

Step 11, Agent_g1 (gen 1), first step:

```
Decision: move to node 0
Reason: "My inherited knowledge says the goal is
```



(a) Efficiency (steps per birth) by condition.

(b) Root (gen-0) vs. child (gen-1+) success rates.

Figure 6: Additional fertility metrics. Novelty 0.7 achieves the best efficiency ratio, and child agents outperform root agents under most strategies.

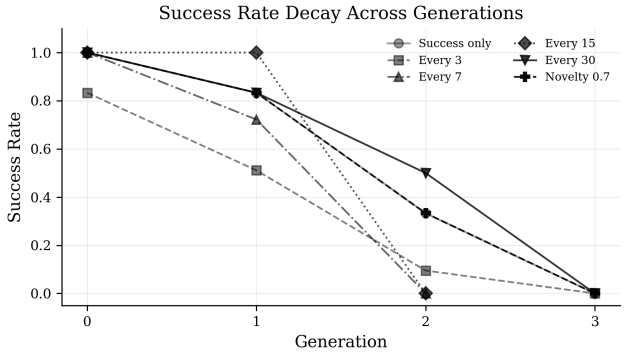


Figure 7: Generational success decay across fertility conditions. Higher-fertility strategies produce deeper lineage trees but with diminishing per-generation success. Novelty 0.7 maintains the best balance between depth and per-generation quality.

```
the red arched door at node 0. Moving directly."
```

The child navigates directly to the goal on its first step using inherited knowledge.

O Repository Structure

```
src/
    config.py          TrialConfig dataclass
    environment.py     RGG, door/signal generation
    agent.py           LLM agent, context, priors
    beliefs.py         Bayesian belief tracking
    reproduction.py    Birth logic, lineage tracking
    runner.py          Trial execution loop
    metrics.py         Signal precision/recall, drift
    skill_library.py   Voyager-inspired conventions
    cloaking.py        Active cloaking (future)
    logger.py          Full transcript logging
experiments/
    a_prior_ablation.py
    b_parent_interaction.py
    c_lexical_shortcuts.py
    e_skill_library.py
    h_cloaked_goals.py
```

```
    i_fertility.py
run_experiments.py     CLI entry point
results/               Output (gitignored)
```